LANE 3 · UNSUPERVISED BEHAVIORAL CLUSTERING

I almost threw away my own capstone lane.

Archetype clustering was supposed to predict underperforming pages. It didn't. Here's the honest result — and why the number that *looked* better was the one to distrust.

CIRCULAR · 0.963 AUC

HONEST · 0.927 AUC

The question

Does knowing a content item's unsupervised **archetype** — a behavioral cluster learned from position, engagement, and content signals — help predict whether it's underperforming, on top of what raw metrics already tell you?

DATASET

~79M

rows · FlyRank warehouse (HF, gated)

WINDOW

1 month

mid-panel, gsc_and_ga4 clients only

What's at stake: which page a human editor reviews first, and for what kind of fix. A wrong call costs a review cycle — every recommendation is reviewed by a person before anything changes.

Built like an audit, not a demo

TARGET	high_performer = avg_ctr > median(avg_ctr) — a within-month proxy, since no ground-truth label exists
FEATURES	12 total: position, impressions, engagement ratios, AI-referral share, age, word count, search volume, competition, content type/intent, + KMeans (k=6) archetype label
BASELINE	expected_ctr_for_position − actual_ctr — the transparent rule an editor already uses
SPLIT	GroupShuffleSplit on client ID, zero overlap confirmed — blocks the model from memorizing client editorial style
LEAKAGE CHECKS	sealed-month leak trap, random vs. grouped split comparison, excluded-field scan

The scoreboard lies if you stop at AUC

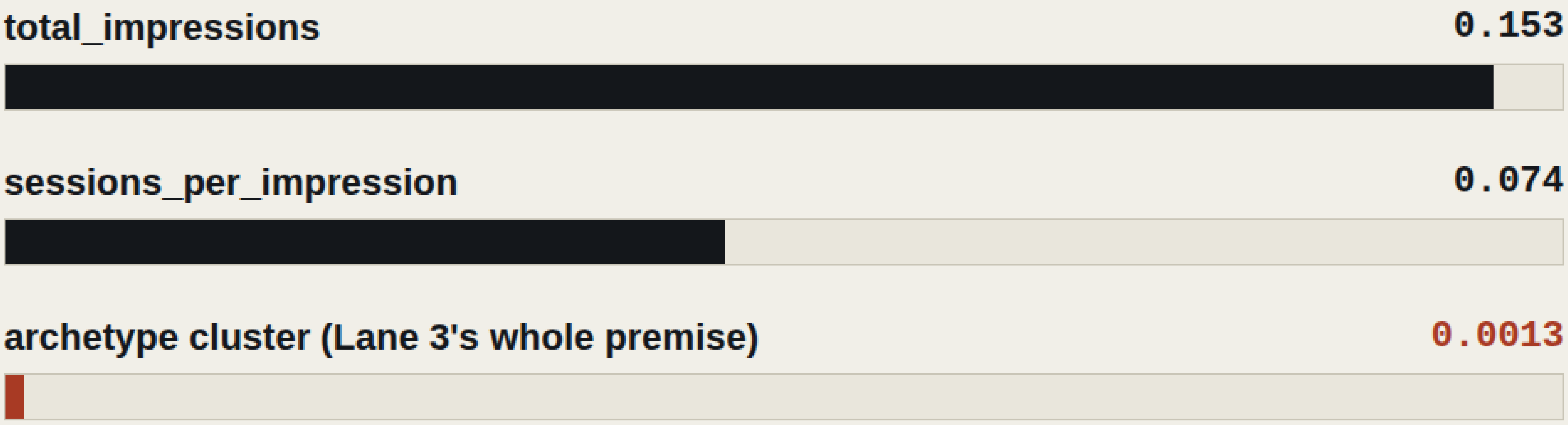
Baseline rule (single signal)	0.963
Random Forest (12 features + archetype)	0.927

CIRCULAR · DISCARD TRUSTWORTHY

`corr(baseline_score, avg_ctr) = −1.000` on the held-out split. The baseline isn't predicting the target — it's a near-exact linear rescaling of it. The model's lower score is the one actually built from independent signal.

What the model actually leans on

Permutation importance, held-out split



Effectively zero. Clustering content into behavioral archetypes did not meaningfully help predict CTR performance in this test.

THE PART WORTH SHARING

A negative result, reported clearly, is still a result.

A big chunk of ML work is knowing when your fancy feature isn't earning its place — and saying so, instead of burying it under a metric that happens to look good.

Limitations, stated plainly

01 Baseline AUC is inflated by near-circularity, not genuine predictive skill.

02 Archetype clustering did not earn a predictive role — a genuine negative result, reported as one.

03 Single cross-sectional month; can't distinguish a consistently weak page from one having a quiet month.

04 No true sealed holdout evaluation — the final month is used only to demonstrate leakage.

05 Every relationship reported is observational — no causal design, no claim of "why."

06 Scoped to gsc_and_ga4-access clients, one month, one client mix.

A ranked queue, not an autopilot

REVIEW_TITLE_META

CTR falls meaningfully below its own position tier's benchmark — the highest-confidence, most reproducible signal in this study.

REFRESH_CANDIDATE

Archetype clusters skewing older and below-median CTR — directionally consistent with FlyRank's portfolio-level refresh findings.

DISTRIBUTE_SUPPORT

Clusters already ranking well with strong CTR — protect and amplify rather than rebuild.

NEVER AUTOMATE

No queue item auto-publishes. Every recommendation requires human review against the no-go list before anything ships.

Full writeup, methodology, notebooks & charts — linked below.

Every number traces back to a re-runnable notebook: data contract, baseline, model + archetype comparison, validation audit, and the action playbook. Random seed fixed throughout.

Grateful to the **FlyRank AI** team for the track and dataset access.